

FlowPOS

A Retail Experience Powered by Computer Vision

Face Recognition · Object Detection · Billing

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What We'll Cover Today

1

The Problem

Why existing systems fall short

2

Our Solution

How FlowPOS works

3

System Architecture

Modules & data flow

4

Enrollment Pipeline

Building a biometric identity

5

Face Recognition

Voting-based identity matching

6

Item Detection

YOLOv8 model training & results

7

Item Tracking & Billing

Tripwire logic & session billing

8

Results & Demo

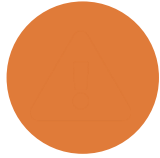
Live system performance

9

Limitations & Next Steps

Honest assessment & roadmap

Traditional Retail Checkout Creates Friction & Loss



Long Queues

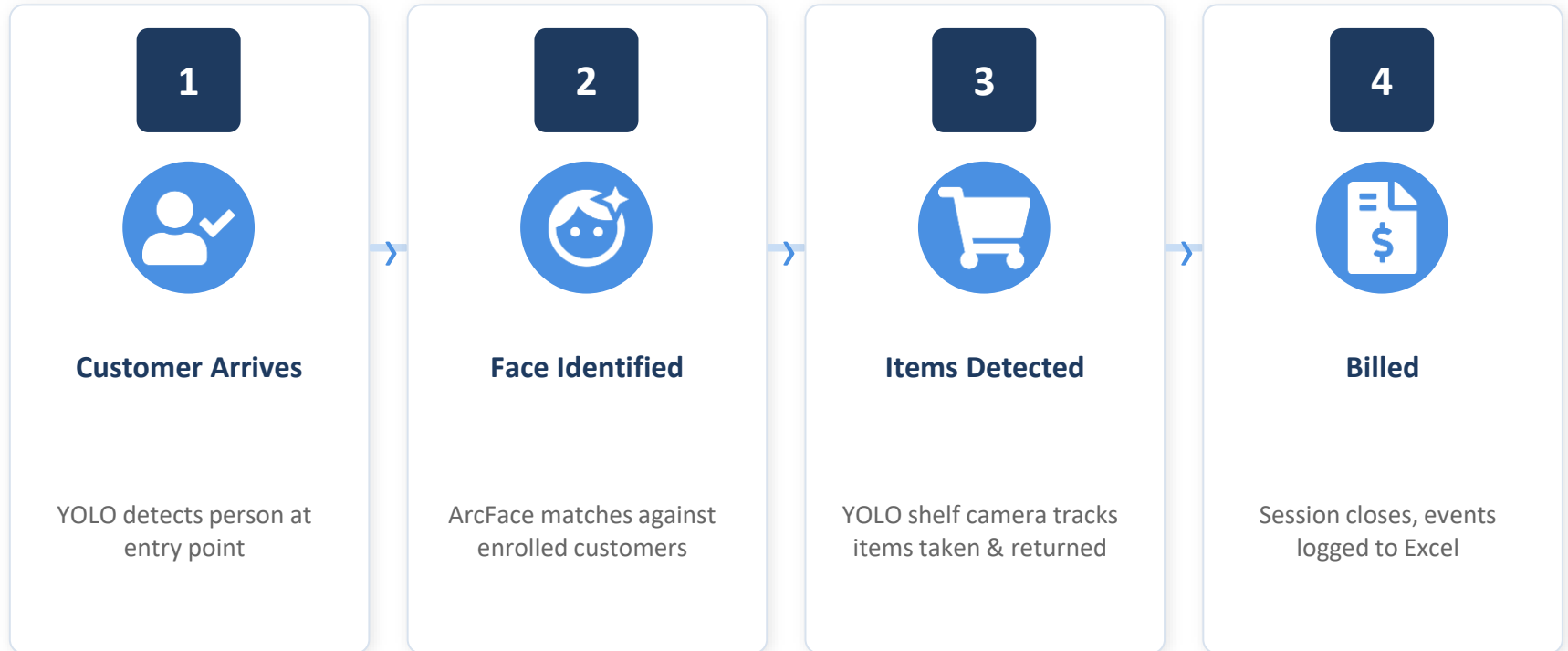
Customers abandon purchases during peak hours, reducing revenue



High Labour Cost

Dedicated checkout staff is expensive and inefficient

FlowPOS: Walk In, Get Recognised, Walk Out



Two-Camera Architecture Separates Concerns Cleanly

Face Recognition

- YOLO — person detection
- MediaPipe — face crop & landmark
- ArcFace — embedding & identity match
- AccessController — fires events

Station
Bridge



Item Detection

- YOLO — shelf item detection
- Tripwire logic — taken / returned
- Billing — session totals
- Excel Logger — receipt output

Enrollment Builds a Biometric Identity in 5 Phases

P1

Camera Capture

Captures frames at different poses: front, left, etc

P2

MediaPipe Detect

Face bounding box detected;

P3

Quality Gate

Blur & brightness checks filter bad frames

P4

CLAHE Preprocess

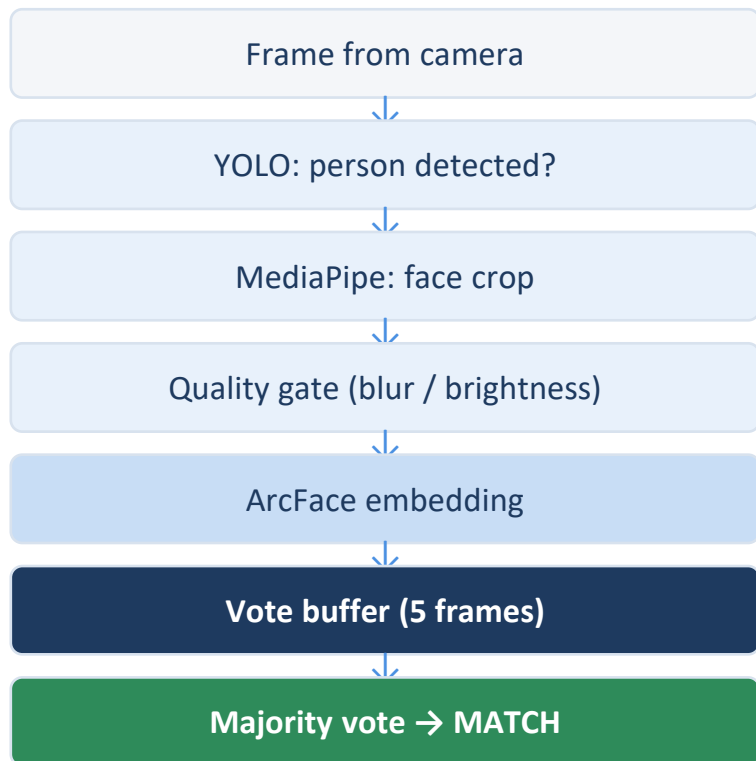
CLAHE equalisation normalises lighting

P5

ArcFace Embed

10 embeddings (512-dim) + centroid stored in SQLite

Recognition Uses Voting to Eliminate False Positives



Key Thresholds

Blur minimum

30 (Laplacian variance)

Brightness range

40 – 250

Cosine threshold

≥ 0.60 to register a vote

Vote confidence floor

≥ 0.60 for final match

Buffer size

5 frames required

Votes to confirm

3 of 5 must agree

Embedding dimension

512-dim (ArcFace)

Item Detection Model

Model Training Details

Architecture	YOLOv8 (Ultralytics)
Task	Object detection — shelf items
Classes	8 item categories
Training epochs	50
Input size	640 × 640 px
Inference	Real-time, camera feed

98.9%

Precision

Almost zero false positives

85.7%

Recall

Strong item detection rate

89.9%

mAP@50

Standard detection benchmark

61.0%

mAP@50–95

Strict multi-IoU benchmark

Tripwire Logic Distinguishes Items Taken from Items Returned

1

Camera Feed

Dedicated second camera watches the items at a fixed angle continuously

2

YOLOv8 Detects Item

Model identifies item class and bounding box on each frame

3

Tripwire Crossed?

Item centroid crosses a pre-set boundary line — 'taken' or 'returned' zone

4

Event Fired

StationBridge notified: item class, direction, and active customer ID

TAKEN

- Item moves from left → right
- Billing session adds item + price

RETURNED

- Item moves from right → left
- Item removed from active total
- No charge applied for returned items

Session Closes and Writes Events to Excel

1

Customer confirmed

AccessController fires CUSTOMER_IDENTIFIED event

2

Session opens

StationBridge creates billing session for the customer

3

Items accumulate

Each TAKEN event adds item + price from price_catalog.py

4

RETURNED reversal

Returned items are deducted from the running total

5

Customer leaves

YOLO loses person detection → session auto-close triggered

6

Receipt written

excel_logger.py appends row to smart_inventory_log.xlsx

What the Excel Receipt Contains

Customer name

From enrollment record

Staff ID

Unique identifier (e.g. SI-001)

Items taken

List of item classes detected

Items returned

Deducted from total

Session total

Final amount charged (¥)

Timestamp

Date and time of session close

Visit count

Cumulative visits for customer

Production-Grade AI Stack Drives Every Component



ArcFace

512-dim face embeddings
Cosine similarity matching
detector_backend: skip



YOLOv8

Person detection (M2)
Shelf item detection (M4)
Custom trained best.pt



MediaPipe

Real-time face detection
Face crop & landmark align
Full-range model



SQLite

Customer biometrics
Embeddings + centroid
Purchase history



TensorFlow

ArcFace model backend
GPU-optional inference
KerasFace weights



OpenCV

Frame capture & resize
CLAHE preprocessing
Laplacian blur scoring

System Identifies Customers & Bills Automatically

What the Live Demo Shows

- Customer enrollment
- System detects customer walking into frame via YOLO — recognition triggers
- ArcFace matches identity within 5-frame voting window and displays name
- Customer picks up item from shelf — item engine logs the event instantly
- Customer leaves — session auto-closes and Excel receipt is written

LIMITATIONS

Four Constraints Define the Current System Boundary



Single-person sessions

System tracks one active customer at a time; crowds cause state conflicts



Lighting sensitivity

Quality gate may reject frames in dim or variable indoor lighting



Windows-only (current)

Camera indices and OpenCV bindings tested only on Windows 11



Enrollment quality dependency

Poor enrollment photos corrupt the centroid and break recognition

Further Improvements

1

Multi-person tracking

Track multiple customers simultaneously using ID assignment per YOLO detection

2

Adaptive quality gate

Auto-calibrate blur threshold to ambient lighting at startup

3

Cloud sync & mobile receipt

Push receipts to customer mobile via cloud — no Excel dependency

4

Active learning enrollment

Continuously improve centroid using validated live-session frames

FlowPOS demonstrates that checkout-free retail is achievable with computer vision.

- ArcFace + MediaPipe delivers robust face recognition
- YOLOv8 enables real-time item-level shelf tracking
- Modular design makes each component independently testable

Thank You — Questions?